# Abhiram Kandiyana

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## SUMMARY

**Ph.D.** candidate in Artificial Intelligence with **4+** years of hands-on experience in **Computer Vision**, **Deep Learning**, **Machine Learning**, and **AI-driven** product development. Proven track record of designing scalable and efficient AI algorithms for automation of **microscopy image analysis** using **Vision Language Models**. Proficient in **Python, OpenCV, Pytorch, Tensorflow, Generative AI, LLM Tuning and data processing**. Passionate about applying AI research to build impactful, real-world products.

## EDUCATION

University of South Florida, **PhD** in Computer Science and Engineering — Jan 2023 – Dec 2026

- GPA: 3.9/4.0
- **Major Area:** Automated Quantification and Explanation of Microscopy Images using Vision Language Models (VLMs)

University of South Florida, **Masters** in Computer Science, Thesis track — Aug 2021 – May 2024

- GPA: 3.9/4.0

## TECHNICAL SKILLS

**Programming**: Python, C, CUDA, SQL

**Frameworks**: PyTorch, Tensorflow OpenCV, Pillow, Keras, Scikit-image, NumPy, Scikit-learn, TorchVision, Matplotlib, LATEX, HuggingFace, Docker, AWS, GIT, vLLM, OpenAI APIs

**AI Assitance**: Chain-Of-Thought Prompting, Few-shot Prompting, ChatGPT (Deep-Research), OpenAI Codex, Cursor, NotebookLM
**Concepts**: Deep Learning, Computer Vision, Image Understanding, Object Detection, image segmentation, NLP, Natural Language Supervision, Distributed Computing, Parameter Efficient Fine-tuning (PEFT) of multi-modal LLMs, Prompt Engineering

## PUBLICATIONS

**Kandiyana, A.**, Mouton, P. R., Kolinko, Y., Hall, L. O., & Goldgof, D. B. (2025). ***Active Prompt Tuning Enables GPT-4o to Do Efficient Classification of Microscopy Images***. IEEE International Symposium on Biomedical Imaging (ISBI).

McCofie, A., **Kandiyana, A.**, Mouton, P. R., Sun, Y., & Goldgof, D. B. (2025). ***Few-Shot Prompting with Vision Language Models for Pain Classification in Infant Cry Sounds.*** IEEE International Symposium on Computer-Based Medical Systems (CBMS).

**Kandiyana, A.**, Mouton, P., Hall, L., & Goldgof, D. (2024). ***Active Prompting of Vision Language Models for Human-in-the-Loop Classification and Explanation of Microscopy Images.*** IEEE Conference on Computer-Based Medical Systems.

**Kandiyana, A.**, Mouton, P. R., Kolinko, Y., Hall, L. O., & Goldgof, D. B. (2024). ***Automatic Classification of Light Microscopy Images Using GPT Models.*** ASEMFL Annual Meeting.

**Kandiyana, A.** (2024). ***Semi-automated Cell Annotation Framework Using Deep Learning.*** Master's Thesis, University of South Florida.

## EXPERIENCE

**Research Assistant**, M-AI Project, University of South Florida | Tampa, FL — Aug 2021 – Present

- Developed a novel Tuning framework (**APT**) for biomedical image understanding, achieving expert-level accuracy (**96%**) while boosting efficiency by up to **96%**, minimizing data and time requirements.
- Fine-tuned foundation VLMs like **GPT-4o, NVLM, CLIP, LLaVA, Llava-Med** to specialize their capabilities for microscopy image classification with an average accuracy of **93%**.
- Engineered automated segmentation solutions integrating heuristic methods and fine-tuned transformer models (Meta SAM), reducing manual annotation time by **57%** while maintaining **> 90%** accuracy.
- Deployed our methods on **AWS** by containerizing them with **Docker**, hosting inference services on **EC2**, and managing model artifacts and datasets using **S3**.

  Hosted VLMs on GPU clusters using **vLLM, PyTorch DDP, Transformers**, making them accessible to anyone in the University with a URL
- Experienced in image pre-processing techniques such as normalization, denoising, color space conversion, histogram

equalization, and augmentation.

**Co-Founder and Founding Engineer**, Mindlr AI | Remote May 2024 – Aug 2025

- Designed and developed an AI search engine from scratch that generates multi-step blueprints for user tasks, and workflows using **LangChain, RAG, PineCone, AWS, Docker, Huggingface and LLMs**
- Bootstrapped the startup from an idea to a beta product within three months while being featured in **BuildSpace**.

**Machine Learning Engineer**, ConceptWaves | Vinshko Technologies, Hyderabad, India Aug 2020 – Aug 2021

- Improved the reliability for online tests by developing a object detection and classification tool as a part of semi-automated proctoring website using Python, PyTorch, TorchVision, OpenCV, and AWS.
- Implemented a computer vision-based grading system using Python, PyTorch and TorchVision, automating the evaluation of handwritten and typed student responses, which reduced grading time by **40%** and provided immediate feedback to learners.
- Optimized energy usage by analyzing historical power data using Machine Learning and Predictive Modeling to identify inefficiencies, reducing energy consumption by **15%**

**Teaching Assistant**, Bellini College, University of South Florida Aug 2021 – Present

- Delivered graduate-level lectures on key **deep learning** topics including data preprocessing, activation functions, regularization techniques, optimization algorithms, backpropagation, and convolutional neural networks (CNNs); independently taught multiple full-length class sessions.
- Designed and led hands-on labs for **Image Processing** covering **OpenCV fundamentals, 3D and stereo imaging**.

## RESEARCH PROJECTS

**Efficient Microscopy Image Classification using VLMs** (Python, Pytorch DDP, Generative AI) Funded by NSF

- Demonstrated generalizability of the approach across diverse datasets, including different brain regions, magnifications and stains, ensuring scalability and adaptability.
- Benchmarking several state-of-the-art VLMs for microscopy image understanding across heterogeneous datasets and biological sub-domains.

## AWARDS, GRANTS AND INVITATIONS

**NIH Entrepreneurship Bootcamp** National Institute of Health

Participated as a key team member in a competitively selected, entrepreneurship and research translation program.
Interviewed 40+ neuro-scientists, researchers across the globe as a part of customer discovery.

**Excellence in Student Research Award** The Corridor 2024

Deep Learning-based Classification of Immunofluorescent-labeled Microglia Cells

**Speaker Invitation** INTERM 2025

Characterization in Biomedical Sciences

**Campus Strategist** Perplexity 2025

Representing Perplexity at University of South Florida

## OTHER PROJECTS

**Error Explainer** (Python, Pytorch, NLP, LLMs, LoRA, COT Prompting)

Developed an AI tool that explains the unintelligible error messages from a compiler. Prompt-tuned and fine-tuned CodeLlama (LLM) on a custom alignment dataset that was crowd-sourced from students at USF achieving a **92%** human evaluation score.

**CUDA-BLAST-2024** (CUDA, Parallel Processing, Dynamic Programming)

Reimplemented Basic Local Alignment Search Tool (BLAST) in CUDA for extremely fast and efficient comparison of protein structures in large-scale databases (100,000 records). Acheived **5x** improvement compared to NCBI BLAST.

**Skin Cancer Classifier** (Python, Tensorflow, CNNs, Computer Vision)

Built and deployed CNN-based models using Python, achieving **78%** top-1 accuracy for skin cancer classification. Performed data augmentation and pre-processing to address class imbalance and improve generalization.